lab 6 Prelab: [illegible] 3/1/6

6.1

Setting up differential amp

$R_{3left} = 2.94\,k\Omega$, $R_{3right} = 2.93\,k\Omega$, $R_{10k} = 9.8\,k\Omega$

$R_{100k} = 100.0\,k, 99.5\,k$, $R_{100} = 99.8, 98.3\,\Omega$

(a)

measured output: $0.91V \rightarrow G_\Delta = 9.1$

expected: $(3000)/(100+138) = 12.3$

no agreement - circuit is not symmetrical and offsets are not the same

(b) $\Delta t = 0 \pm 4\,\mu s$, $\rightarrow \Delta\theta = 0 \pm 1.4°$, agrees w/ predicted since $V_{out-} = -V_{out+} = -V_{in+}$ for an ideal system

(c) $V_{in} = 1V$ amplitude

$\hookrightarrow V_{out} = 0.1465 \pm 0.0003V \rightarrow G_c = 0.1465 \pm 0.0003$

expected $= 3k\Omega/(2 \cdot 10k\Omega + 138\Omega + 100\Omega) = 0.1480 \pm 0.002 \rightarrow$ agrees!

$CMRR = 20\log_{10}(G_\Delta/G_c) = 38.05 \pm 0.24\,dB$

6.2

(a) $V_{\Delta out} = 1.171 \pm 0.032V$, $V_{cout} = 0.8 \pm 0.007\,mV$

$G_\Delta = 11.705$ $\quad G_c = 0.0008 \pm 0.000007$

$\hookrightarrow CMRR = 83.31 \pm 0.25\,dB$

VS 03/06

The current source is much more stiff and its $r_{eff} \gg 10k\Omega$, without squashing current - it behaves more like an ideal current source

(b) when one is warmed, the noise on one channel increases - the channel associated with the transistor being warmed. However, this is possibly due to movement. when both are warmed, the signal is much more noisy

(c) with unmatched JFETs, the voltage offset is significantly worse

6.3

(a) (b) built to the best of our ability

(c) screenshot in full lab report

(d)

resonant frequency: 796.0 ± 0.05 kHz , $f_{3dB} = 779.8 \pm 0.1$, 813.6 ± 0.1 kHz

$L = (2\pi f_{res})^2 C = 14.7 \pm 0.4$ mH $\quad \Delta f_{3dB} = 33.8 \pm 0.1$ kHz

Primary sources of error: bad winding, noise/extra capacitance in the circuit

(e) $Q = 23.55 \pm 0.1$

(a) $R_p \geq (Q\sqrt{L/C}) = 1.73 \pm 0.05$ kΩ , not plausible

(b) $R_s \leq (\sqrt{L/C}/Q) = 3.12\ \Omega$

$R_{inductor} = 0.174\ \Omega$, not a great source

$R_{signal} = R_s \rightarrow k = 0.08$, implausible $\rightarrow$ must be dissipation across C

(f) $f_{res} = 1.326 \pm 0.01$ MHz

expected: 1.354 ± 0.03 MHz - agrees!

(g)

$G_{ain} = 1.0297/0.0974 = 10.58 \pm 0.01$ at f_{res}

6.4

(a) built

(b) see pdf, rectification is visible

(c) modulation setup

(d) kinda matches? expected peaks are not super tall

(e) screenshot in final submission

$f_{3dB} = 7.23\ kHz$, $G_{exp} = 5.7 \cdot 10^{-3}$, measured 1e-2

(f) $f_{3dB} = 165 \pm 8\ Hz$, only cuts in slightly

(g)

	f (Hz)	G measured	G predicted	agree?
below (f) filter	10	0.047 ± 0.003	3.135	✗
	50	0.7353 ± 0.06	3.135	✗
	200	0.7252 ± 0.388	3.135	✗
audio band	700	3.199 ± 0.07	3.132	✓
	1000	3.111 ± 0.065	3.129	✓
above audio band	10000	1.816 ± 0.040	2.655	✗
	16000	1.668 ± 0.020	2.24	✗
	32000	0.1543 ± 0.113	1.396	✗

low values due to (f), high due to low-pass ifet amp

6.5

(a) finished constructing the circuit

(b) max $V = \pm 6V$ (not ~ $\pm 12V$ as expected), for V_{out2} we only get $\pm 100mV$

(c) tone is audible!

(d) tuned to AM 1400 - super audible

(e)

Capacitance	Predicted resonance ($\pm \sim 20kHz$)	station
0.87nf	1410 kHz	KVTO (Sing Tao)
2.74nf	795 kHz	KSFO (conservative talk)
5.09uf	583 kHz	KEAR (family radio)
1.8nF	980 kHz	KNEW (talk)
2.7nF	800 kHz	same as above but mixed with Sing Tao

6.5 [illegible] 3/12/26

(f) clipping in pdf, distortion audible

6.6

(a) built!

(b) See pdf

(c) largest linear signal (input) 1.3V amplitude triangle wave

6.7

(a) no distortion visible at 5V - much better!

(b) max attenuation: $1 - \frac{9.5V}{2} / \frac{436mV}{2} = 1 - 0.04550 = 0.9545$

$$r_{ds} = (R_S)(1/G_{max} - 1) = 20.5 \pm 0.2\ k\Omega$$

6.8

(a) The DC offset is regulated by the pot, while the AC component is due to the signal input

(b) built!

(c) in full pdf

6.9

(a) $8.13 \pm 0.3\ nF \leftrightarrow 461 \pm 10$ kHz resonant frequency

network analysis:

$f_{res} = 489 \pm 2$ kHz $\leftrightarrow C = 7.2 \pm 0.2$ nF

(b) built!

(c) audible tone!!

(d) pitch shifting works!

(e) detuning works!

6.9

(f) played a jazz tune, got checked off

Celeste Vun
6.9 3/13/24